

Homework 3

ASEN 5335 Aerospace Environment

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October 6, 2026

Problem 1: Plotting the Ionosphere

(i.)

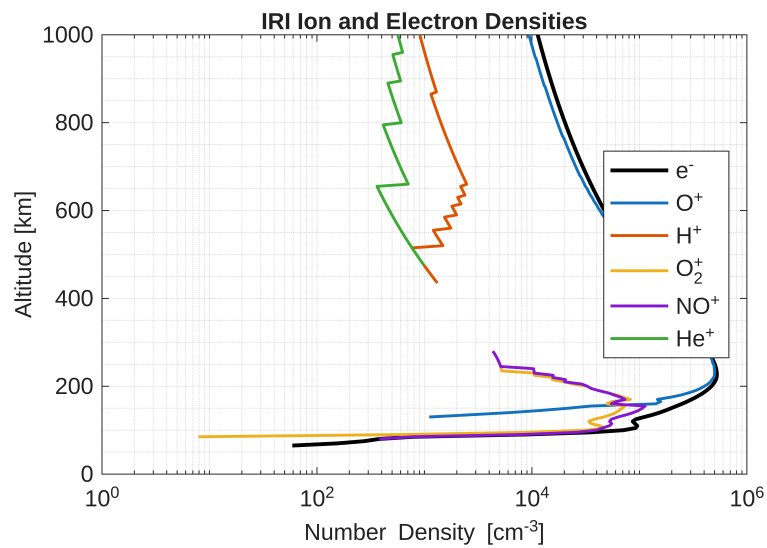


Figure 1: IRI Output

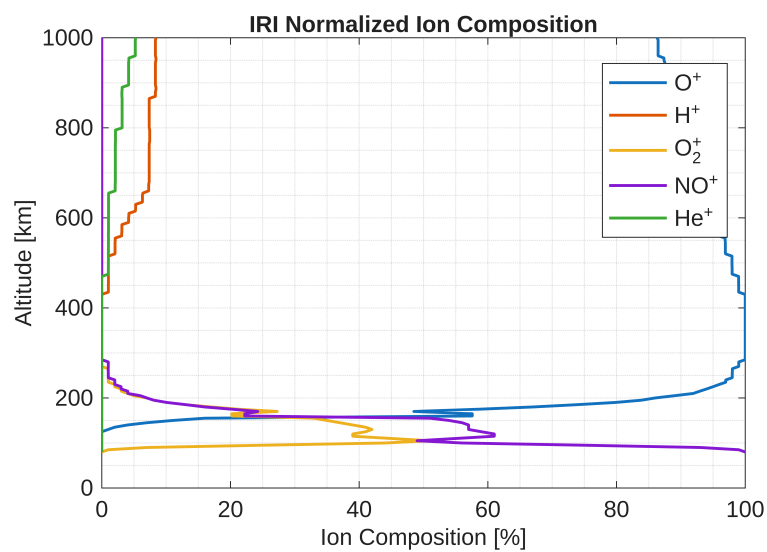


Figure 2: IRI Output Normalized

(ii.)

Below 500 km, we can observe that NO^+ dominates between 0-160 km. O_2^+ contributes more than 20% at between 90 and 175 km.

(iii.)

By convention 1 TECU is 10^{16} el/m². Integrating the total electrons from the IRI output, we find the TEC of the entire IRI output to be 12.47 TECU. When the peak electron density is 525696 m^{-3} , we can integrate the topside and bottomside separately to find they are 3.55 TECU and 8.92 TECU respectively. Normalizing against the total TEC found, we find they are 28.49% and 71.51% respectively.

Problem 2: Peak Ionization

(i.)

E Region

We are able to use the Chapman relation equation,

$$n_e(z, \theta) = \left(\frac{q_{max}}{\alpha_{eff}} \right)^{1/2} \exp \left[\frac{1}{2} (1 - z_1 - e^{-z_1}) \right]$$

Solving for q_{max} ,

$$q_{max} = \alpha_{eff} n_e^2 \exp[-(1 - z_1 - e^{-z_1})]$$

At the peak altitude $z = z_{max}$, $z_1 = 0$, so the exponential term is 1 and $n_e = n_{e,max}$:

$$q_{max} = \alpha_{eff} n_{e,max}^2$$

$$q_{max_E} = (10^{-8})(1.5 \times 10^5)^2$$

$$\boxed{q_{max_E} = 225 \text{ cm}^{-3} \text{ s}^{-1}}$$

F1 Region

Repeat the calculation above for the F1 Region,

$$q_{max_{F1}} = (7 \times 10^{-9})(2.5 \times 10^5)^2$$

$$\boxed{q_{max_{F1}} = 437.5 \text{ cm}^{-3} \text{ s}^{-1}}$$

F2 Region

Repeat the calculation above for the F2 Region,

$$q_{max_{F2}} = (5 \times 10^{-9})(5.0 \times 10^5)^2$$

$$\boxed{q_{max_{F2}} = 1250 \text{ cm}^{-3} \text{ s}^{-1}}$$

(ii.)

The reason why the ionization peak altitude z_{max} moves upwards with increasing solar zenith angle χ when all other factors are kept constant is because when examining the equation,

Problem 3: Spacecraft Charging in LEO

We are able to use the charging equation,

$$\frac{dQ}{dt} = I_e + I_i + I_{SEE,i} + I_b + I_{ph}$$

By assuming $\frac{dQ}{dt} = 0$, the eclipse term, $I_{ph} = 0$, and no secondary electron emission or backscattered electron current, $I_{SEE,i} = I_b = 0$, the equation simplifies to,

$$0 = I_e + I_i$$

Using equations for I_i , and I_e ,

$$I_i = qn_i v_0 A_i$$

$$I_e = -\frac{1}{4} q n_e v_{e,th} A_e e^{qV/kT_e}$$

and calculating

$$V_{fl} = \frac{k_B T_e}{q} \ln \left(\frac{4v_0 A_i}{v_{e,th} A_e} \right)$$

The thermal voltage at a constant $T_e = 1000$ K is,

$$\frac{k_B T_e}{q} = \frac{(1.381 \times 10^{-23} \text{ J/K})(1000 \text{ K})}{1.602 \times 10^{-19} \text{ C}} = 0.0862 \text{ V}$$

The orbital velocity at 400 km altitude is,

$$v_0 = \sqrt{\frac{\mu}{R_E + h}} = \sqrt{\frac{3.986 \times 10^{14} \text{ m}^3/\text{s}^2}{6.378 \times 10^6 \text{ m} + 4.0 \times 10^5 \text{ m}}} = 7.67 \text{ km/s}$$

The electron thermal velocity is,

$$v_{e,th} = \sqrt{\frac{3k_B T_e}{m_e}} = \sqrt{\frac{3(1.381 \times 10^{-23} \text{ J/K})(1000 \text{ K})}{9.109 \times 10^{-31} \text{ kg}}} = 213 \text{ km/s}$$

If this is a 6U Cubesat, ions are collected only on the 20×30 cm ram face, electrons collect over entire surface,

$$A_i = (0.2 \text{ m})(0.3 \text{ m}) = 0.06 \text{ m}^2$$

$$A_e = 2(0.02 \text{ m}^2 + 0.03 \text{ m}^2 + 0.06 \text{ m}^2) = 0.22 \text{ m}^2$$

Plug into the floating potential equation,

$$\frac{4v_0A_i}{v_{e,th}A_e} = \frac{4(7.67 \text{ km/s})(0.06 \text{ m}^2)}{(213 \text{ km/s})(0.22 \text{ m}^2)} = 0.0392$$

$$V_{fl} = (0.0862 \text{ V}) \ln(0.0392) = (0.0862 \text{ V})(-3.24)$$

$V_{fl} \approx -0.28 \text{ V}$

My assumptions were equilibrium ($\frac{dQ}{dt} = 0$), eclipse ($I_{ph} = 0$), no secondary or backscattered electrons, $n_e = n_i$, ions collected only on the ram face, electrons collected on the entire surface, and a $10 \times 20 \times 30$ cm CubeSat in a circular orbit. We assume that the spacecraft is spherical in order to use the equations however it should be noted that we still use the area of the 6U CubeSat provided.

Problem 4: Spacecraft Charging in GEO

Use the charging equation,

$$\frac{dQ}{dt} = I_e + I_i + I_{SEE,i} + I_b + I_{ph}$$

Assume $\frac{dQ}{dt} = 0$ and $I_{SEE,i} = I_b = 0$,

$$0 = I_e + I_i + I_{ph}$$

The electron current is,

$$I_e = -\frac{1}{4}qn_e v_{e,th} A_e e^{qV/k_B T_e} \quad \text{for } V < 0$$

$$I_e = -\frac{1}{4}qn_e v_{e,th} A_e \left(1 + \frac{qV}{k_B T_e}\right) \quad \text{for } V > 0$$

The ion current is,

$$I_i = \frac{1}{4}qn_i v_{i,th} A_i e^{-qV/k_B T_i} \quad \text{for } V > 0$$

$$I_i = \frac{1}{4}qn_i v_{i,th} A_i \left(1 - \frac{qV}{k_B T_i}\right) \quad \text{for } V < 0$$

The photocurrent from a sunlit surface is,

$$I_{ph} = \left(40 \frac{\mu\text{A}}{\text{m}^2}\right) A \cos \theta e^{-qV/k_B T_{ph}}, \quad k_B T_{ph} \simeq 2 \text{ eV}$$

$n_e = n_i = 0.5 \text{ cm}^{-3}$, $T_e = T_i = 5 \times 10^6 \text{ K}$, and $A_i = A_e = A$ so the area cancels. The thermal voltage and thermal velocities are,

$$\frac{k_B T}{q} = \frac{(1.381 \times 10^{-23} \text{ J/K})(5 \times 10^6 \text{ K})}{1.602 \times 10^{-19} \text{ C}} = 431 \text{ V}$$

$$v_{e,th} = \sqrt{\frac{3k_B T_e}{m_e}} = 1.51 \times 10^4 \text{ km/s}$$

$$v_{i,th} = \sqrt{\frac{3k_B T_i}{m_i}} = 352 \text{ km/s}$$

Shadow Surface

In shadow $I_{ph} = 0$ and $V < 0$, the electron and ion currents for $V < 0$,

$$0 = -\frac{1}{4}qn_e v_{e,th} e^{qV/k_B T_e} + \frac{1}{4}qn_i v_{i,th} \left(1 - \frac{qV}{k_B T_i}\right)$$

Solving numerically with MATLAB,

$$\boxed{V_{shadow} \approx -1079 \text{ V}}$$

Sunlit Surface

In sunlight $V > 0$, and assuming $\cos \theta = 1$ and $k_B T_{ph} \simeq 2 \text{ eV}$,

$$0 = -\frac{1}{4} q n_e v_{e,th} \left(1 + \frac{qV}{k_B T_e} \right) + \frac{1}{4} q n_i v_{i,th} e^{-qV/k_B T_i} + \left(40 \frac{\mu\text{A}}{\text{m}^2} \right) e^{-qV/k_B T_{ph}}$$

Solving numerically with MATLAB,

$$\boxed{V_{sun} \approx 9.8 \text{ V}}$$

Potential Difference

$$\Delta V = V_{sun} - V_{shadow} = 9.8 \text{ V} - (-1079 \text{ V})$$

$$\boxed{\Delta V \approx 1089 \text{ V}}$$

Problem 5: Radio Wave Propagation

(i.)

To find the minimum radio frequency to communicate from ground to space, we would need the frequency that exceeds the largest critical frequency in the atmosphere. The critical frequency (with n_e in cm^{-3}) is,

$$f_c = 9 \times 10^{-3} \sqrt{n_e} \text{ MHz}$$

$$f_c = 6.53 \text{ MHz}$$

(ii.)

To find the region where this frequency reflect, we need to find the region where radio waves are below this frequency. This would be below 225 km. Which would be the F2 region.

(iii.)

To find the region where at 2 MHz signal would reflect, we would treat that frequency as the critical frequency to find in the IRI output. This would occur at about 97 km, which would be the D region.

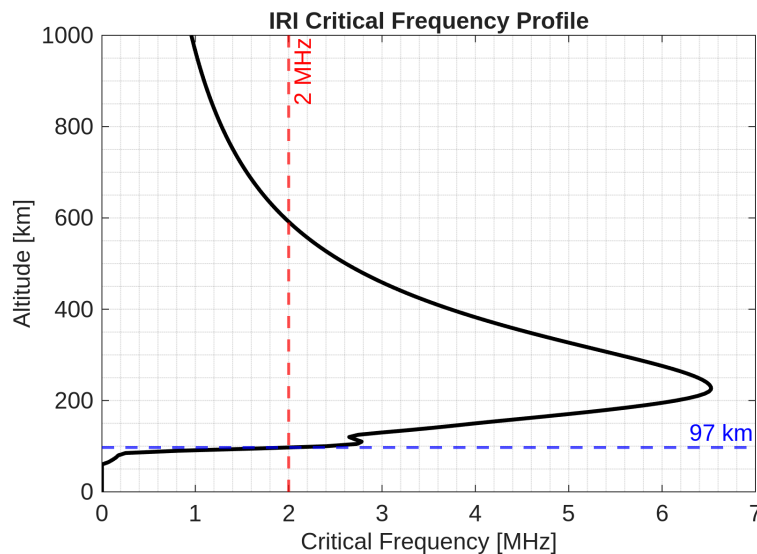


Figure 3: IRI Critical Frequency

Problem 6: Radio Blackout

(i).

The reason that the spacecraft temporarily loses the ability to communicate with the ground is because during re-entry, there is an envelope of ionized atmosphere around the spacecraft, due to the re-entry processes' heat generated. This envelope of ionized atmosphere would have some plasma frequency that would cause radio signals to reflect, causing this radio blackout.

(ii).

If the module communicates with the ground at 2282.5 MHz, to find the electron density in the plasma sheath region, we use the equation,

$$\begin{aligned}
 f_c &= 9 \times 10^{-3} \sqrt{n_{ec}} \\
 &\text{rewrite for } n_{ec}, \\
 n_{ec} &= \left(\frac{f_c}{9 \times 10^{-3}} \right)^2 \\
 n_{ec} &= \left(\frac{2282.5}{9 \times 10^{-3}} \right)^2 = 6.432 \times 10^{10} \text{ cm}^{-3} \\
 &\boxed{n_{ec} = 6.432 \times 10^{16} \text{ m}^{-3}}
 \end{aligned}$$

Problem 7: GPS Delays

(i).

We seek the path length delay through the ionosphere for 10 TECU using, $L1 = 1575.42$ MHz and $L2 = 1227.60$ MHz, we can use the equation,

$$\Delta l_{iono} = -\frac{40.3}{f^2} TEC$$

$$\text{where } TEC = 10 \text{ TECU} = (10)(10^{16} \text{ el/m}^2) = 10^{17} \text{ el/m}^2$$

$$\Delta l_{ionoL1} = -\frac{(40.3)(10^{17} \text{ el/m}^2)}{(1575.42 \times 10^6 \text{ Hz})^2}$$

$$\boxed{\Delta l_{ionoL1} = -1.62 \text{ m}}$$

$$\Delta l_{ionoL2} = -\frac{(40.3)(10^{17} \text{ el/m}^2)}{(1227.60 \times 10^6 \text{ Hz})^2}$$

$$\boxed{\Delta l_{ionoL2} = -2.67 \text{ m}}$$

(ii).

We are able to use multiple frequencies because of the fact that ionospheric delay depends on frequencies. By modeling two frequencies provided by GPS satellites, we can then account for these ionospheric errors. By reducing this increased path due to the ionosphere, we can improve our pseudoranging and have a better position estimate.

Problem 8: D Region Signal Attenuation

The electron-neutral collision frequency is found from the MSIS neutral density and the IRI electron temperature,

$$\nu_{en} = 5.4 \times 10^{-10} n_n T_e^{1/2} \text{ s}^{-1}$$

The electron gyrofrequency for $|B_0| = 50,000 \text{ nT}$ is,

$$\omega_c = \frac{q_e B_0}{m_e} = 8.79 \times 10^6 \text{ rad/s} \quad (f_c = 1.40 \text{ MHz})$$

The differential absorption (with n_e in m^{-3}) for both the $+$ and $-$ solutions is,

$$\frac{dA}{dl} = 4.6 \times 10^{-5} \frac{n_e \nu}{\nu^2 + (\omega \pm \omega_c)^2} \text{ dB/m}$$

Integrate with MATLAB through the IRI profile with $f = 10 \text{ MHz}$,

$$A_+ \approx 1.2 \text{ dB}$$

$$A_- \approx 2.1 \text{ dB}$$

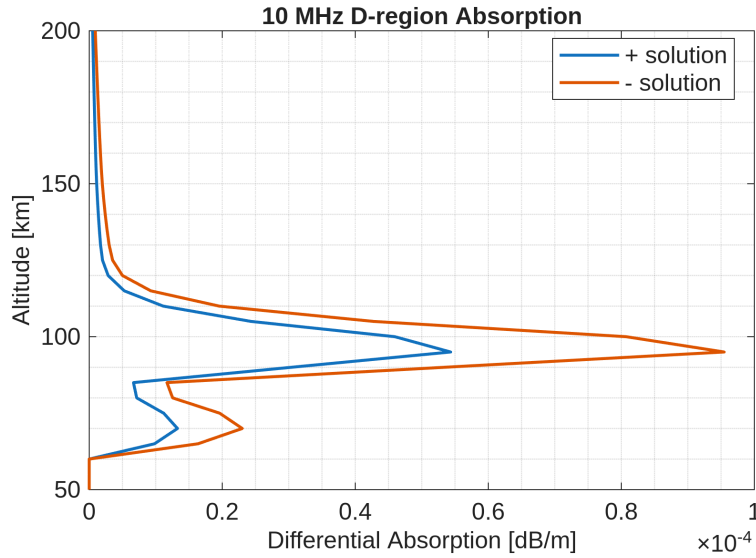


Figure 4: 10 MHz Differential Absorption

Problem 9: ICON Mission

My favorite instrument aboard ICON would be MIGHTI, built by the Naval Research Laboratory. Its purpose is to measure the neutral winds and temperatures of the thermosphere between roughly 90 and 300 km. It's made up of two identical sensors, MIGHTI-A and MIGHTI-B, that look at the Earth with fields of view perpendicular to each other. Its SWAP is spread across two sensors, a shared camera electronics box, and a calibration lamp assembly, but each sensor's baffle is about 62 cm long, the cameras use roughly 11 to 13 watts, and the calibration lamps weigh about 2.7 kg. It works by observing the red and green airglow lines of atomic oxygen, which are Doppler shifted when the neutral gas is moving, and measuring that shift with an interferometer gives the wind along the line of sight. Since MIGHTI-B views the same patch of atmosphere minutes after MIGHTI-A, the two views are combined into the full horizontal wind vector. The instrument contributes to ICON's goal of connecting the lower atmosphere to the ionosphere by measuring the neutral winds that occur, since these winds generate electric fields through the ionosphere and push plasma along the field lines, which can then be compared with the plasma motion measured by IVM.

MATLAB Code

```

1  %% ASEN 5335 Aerospace Environment
2  %% Justin Le
3  %% HW3 Problem 1
4
5  clc;clear; close all
6
7  % Read IRI output
8  data = readmatrix('iri_output.txt', 'FileType', 'text', 'NumHeaderLines', 30);
9  data(data == -1) = NaN; % IRI uses -1 for missing values
10
11  alt = data(:,1); % km
12  Ne = data(:,2); % cm-3
13  pct = data(:,7:12); % ion percentages of Ne: O+ N+ H+ He+ O2+ NO+
14
15  % Set ion number densities [cm-3]
16  n_ion = Ne .* pct / 100;
17  n_Op = n_ion(:,1);
18  n_Hp = n_ion(:,3);
19  n_Hep = n_ion(:,4);
20  n_O2p = n_ion(:,5);
21  n_NOp = n_ion(:,6);
22
23  % Consistent ion colors across figures: O+ H+ O2+ NO+ He+
24  c = lines(5);
25
26  % Plotting
27  figure;
28  semilogx(Ne, alt, 'k-', 'LineWidth', 2); hold on
29  semilogx(n_Op, alt, 'Color', c(1,:), 'LineWidth', 1.5);
30  semilogx(n_Hp, alt, 'Color', c(2,:), 'LineWidth', 1.5);
31  semilogx(n_O2p, alt, 'Color', c(3,:), 'LineWidth', 1.5);
32  semilogx(n_NOp, alt, 'Color', c(4,:), 'LineWidth', 1.5);
33  semilogx(n_Hep, alt, 'Color', c(5,:), 'LineWidth', 1.5);
34  grid minor
35  xlabel('Number Density [cm-3]')
36  ylabel('Altitude [km]')
37  title('IRI Ion and Electron Densities')
38  legend('e-', 'O+', 'H+', 'O2+', 'NO+', 'He+', 'Location', 'best')
39  set(gcf, 'Units', 'inches', 'Position', [0 0 6 4]);
40  set(findall(gcf, '-property', 'FontSize'), 'FontSize', 12);
41  exportgraphics(gcf, fullfile('figures', 'HW3_P1_iri_output.png'), 'Resolution', 900);
42
43  % Normalized as percentages
44  n_plot = [n_Op n_Hp n_O2p n_NOp n_Hep];
45  pct_norm = 100 * n_plot ./ sum(n_plot, 2, 'omitnan');
46
47  figure;
48  h = plot(pct_norm, alt, 'LineWidth', 1.5);
49  set(h, {'Color'}, num2cell(c, 2));
50  grid minor
51  xlim([0 100])
52  xlabel('Ion Composition [%]')
53  ylabel('Altitude [km]')
54  title('IRI Normalized Ion Composition')
55  legend('O+', 'H+', 'O2+', 'NO+', 'He+', 'Location', 'best')
56  set(gcf, 'Units', 'inches', 'Position', [0 0 6 4]);

```

```
57 set(findall(gcf, '-property', 'FontSize'), 'FontSize', 12);
58 exportgraphics(gcf, fullfile('figures', 'HW3_P1_ion_composition.png'), 'Resolution', 900);
59
60 % Total electron content (TEC), 1 TECU = 1e16 el/m^2
61 TECU = 1e16;
62 z_m = alt * 1e3; % m
63 Ne_m = Ne * 1e6; % m^-3
64 Ne_m(isnan(Ne_m)) = 0; % no electrons where IRI has no data
65
66 [NmF2, k_pk] = max(Ne); % peak electron density splits topside/bottomside
67 hmF2 = alt(k_pk);
68
69 TEC_total = trapz(z_m, Ne_m) / TECU
70 TEC_bottom = trapz(z_m(1:k_pk), Ne_m(1:k_pk)) / TECU
71 TEC_top = trapz(z_m(k_pk:end), Ne_m(k_pk:end)) / TECU
72
73 100*(TEC_bottom/TEC_total)
74 100*(TEC_top/TEC_total)
```

```

1  %% ASEN 5335 Aerospace Environment
2  %% Justin Le
3  %% HW3 Problem 4
4
5  clc; clear; close all
6
7  % Constants
8  qe = 1.602176634e-19; % C
9  kB = 1.380649e-23; % J/K
10 me = 9.1093837e-31; % kg
11 mi = 1.67262192e-27; % kg, H+ at GEO
12
13 % GEO plasma
14 n = 0.5e6; % cm^-3 -> m^-3
15 T = 5e6; % K, Te = Ti
16 kT = kB * T / qe; % thermal voltage [V]
17 fprintf('kT/q = %.1f V\n', kT);
18
19 % Thermal velocities [m/s]
20 ve = sqrt(3 * kB * T / me);
21 vi = sqrt(3 * kB * T / mi);
22
23 % Photocurrent (Eq. 28), surface normal to sun
24 Jph0 = 40e-6; % A/m^2
25 kTph = 2; % eV
26
27 % Random current densities at V = 0 if areas cancel out
28 Je0 = 0.25*qe*n*ve;
29 Ji0 = 0.25*qe*n*vi;
30
31 % Shadow surface: dQ/dt = Ie + Ii = 0, V < 0
32 V_neg = linspace(-10*kT, 0, 1e5); % V
33 dQ_shadow = -Je0*exp(V_neg/kT) + Ji0*(1 - V_neg/kT);
34 k = find(dQ_shadow < 0, 1, 'first'); % first sign change in net current
35 V_shadow = interp1(dQ_shadow(k-1:k), V_neg(k-1:k), 0);
36 fprintf('Shadow potential = %.1f V (%.2f kTe)\n', V_shadow, V_shadow/kT);
37
38 % Sunlit surface: dQ/dt = Ie + Ii + Iph = 0, V > 0
39 V_pos = linspace(0, 100, 1e5); % V
40 dQ_sun = -Je0*(1 + V_pos/kT) + Ji0*exp(-V_pos/kT) + Jph0*exp(-V_pos/kTph);
41 k = find(dQ_sun < 0, 1, 'first'); % first sign change in net current
42 V_sun = interp1(dQ_sun(k-1:k), V_pos(k-1:k), 0);
43 fprintf('Sunlit potential = %.2f V\n', V_sun);
44
45 % Differential charging
46 dV = V_sun - V_shadow;
47 fprintf('Potential difference = %.1f V\n', dV);

```

```

1  %% ASEN 5335 Aerospace Environment
2  %% Justin Le
3  %% HW3 Problem 5
4
5  clc; clear; close all
6
7  % Read IRI output
8  data = readmatrix('iri_output.txt', 'FileType', 'text', 'NumHeaderLines', 30);
9  data(data == -1) = NaN; % IRI uses -1 for missing values
10
11  alt = data(:,1); % km
12  Ne = data(:,2); % cm-3
13  Ne(isnan(Ne)) = 0;
14
15  % Critical frequency profile [MHz], Module 3 Eq. 31 ( $n_e$  in cm-3)
16  fc = 9e-3 * sqrt(Ne);
17
18  % (i) Critical frequency foF2 = minimum frequency to reach space
19  [foF2, k_pk] = max(fc);
20  hmF2 = alt(k_pk);
21  fprintf('foF2 = %.2f MHz at %.0f km\n', foF2, hmF2);
22
23  % (iii) Reflection altitude of a 2 MHz wave: first altitude where  $fc = f$ 
24  f = 2; % MHz
25  k = find(fc >= f, 1, 'first');
26  z_reflect = interp1(fc(k-1:k), alt(k-1:k), f);
27  fprintf('2 MHz reflects at %.1f km\n', z_reflect);
28
29  % Plot critical frequency profile
30  figure;
31  plot(fc, alt, 'k-', 'LineWidth', 2); hold on
32  xline(f, 'r--', '2 MHz', 'LineWidth', 1.5);
33  yline(z_reflect, 'b--', sprintf('%.0f km', z_reflect), 'LineWidth', 1.5);
34  grid minor
35  xlabel('Critical Frequency [MHz]')
36  ylabel('Altitude [km]')
37  title('IRI Critical Frequency Profile')
38  set(gcf, 'Units', 'inches', 'Position', [0 0 6 4]);
39  set(findall(gcf, '-property', 'FontSize'), 'FontSize', 12);
40  exportgraphics(gcf, fullfile('figures', 'HW3_P5_plasma_freq.png'), 'Resolution', 300)

```

```

1  %% ASEN 5335 Aerospace Environment
2  %% Justin Le
3  %% HW3 Problem 8
4
5  clc; clear; close all
6  addpath(fullfile('..', 'HW2')) % MSISatmosphere1000.m
7
8  % Read IRI output
9  data = readmatrix('iri_output.txt', 'FileType', 'text', 'NumHeaderLines', 30);
10 data(data == -1) = NaN; % IRI uses -1 for missing values
11
12 alt = data(:,1); % km
13 Ne = data(:,2) * 1e6; % Convert cm-3 to m-3
14 Te = data(:,6); % K
15 Ne(isnan(Ne)) = 0; % no electrons where IRI has no data
16
17 % Neutral densities from MSIS [cm-3]
18 msis = MSISatmosphere1000(alt);
19 n_n = msis.n2 + msis.o2 + msis.o;
20
21 % Electron-neutral collision frequency [s-1]
22 nu = 5.4e-10 * n_n .* sqrt(Te);
23 nu(isnan(nu)) = 0;
24
25 % Constants
26 qe = 1.602176634e-19; % C
27 me = 9.1093837e-31; % kg
28 B0 = 50000e-9; % T
29
30 % Wave and gyro frequencies [rad/s]
31 f = 10e6; % Hz
32 w = 2*pi*f;
33 w_c = qe * B0 / me;
34 fprintf('f_H = %.2f MHz\n', w_c/(2*pi*1e6));
35
36 % Differential absorption [dB/m] for + (w + w_c) and - (w - w_c) solutions
37 dAdl_plus = 4.6e-5 * Ne .* nu ./ (nu.^2 + (w + w_c)^2);
38 dAdl_minus = 4.6e-5 * Ne .* nu ./ (nu.^2 + (w - w_c)^2);
39
40 % Total attenuation along vertical path [dB]
41 z_m = alt * 1e3; % m
42 A_plus = trapz(z_m, dAdl_plus);
43 A_minus = trapz(z_m, dAdl_minus);
44 fprintf('A+ attenuation = %.2f dB\n', A_plus);
45 fprintf('A- attenuation = %.2f dB\n', A_minus);
46
47 % Plot differential absorption profile
48 figure;
49 plot(dAdl_plus, alt, 'LineWidth', 1.5); hold on
50 plot(dAdl_minus, alt, 'LineWidth', 1.5);
51 grid minor
52 ylim([50 200])
53 xlabel('Differential Absorption [dB/m]')
54 ylabel('Altitude [km]')
55 title('10 MHz D-region Absorption')
56 legend('+ solution', '- solution', 'Location', 'best')
57 set(gcf, 'Units', 'inches', 'Position', [0 0 6 4]);
58 set(findall(gcf, '-property', 'FontSize'), 'FontSize', 12);
59 exportgraphics(gcf, fullfile('figures', 'HW3_P8_absorption.png'), 'Resolution', 300)

```